

A Packing of 10 Four-Spaces in $\mathbb{R}^9$

Limin Ge*

October 9, 2026

Abstract

We give 10 4-dimensional subspaces of $\mathbb{R}^9$ whose smallest squared chordal distance is at least 2.468863. This cell of the table of best Grassmannian packings has no published value. The best value implied by known packings is 2.444132, obtained by deleting 1 of the 11 four-spaces of the packing found by this search. The value is checked with exact rational arithmetic from integer coordinates. We do not claim it is optimal.

1 Result

Theorem. *There are 10 subspaces of dimension 4 in $\mathbb{R}^9$ whose pairwise squared chordal distances $D(a, b) = 4 - \text{tr}(P_a P_b)$ are all at least*

$$D_{\min} = 2.468863 \dots$$

The exact value of $D_{\min}$, a rational number, is given in Appendix A.

2 Context

N. J. A. Sloane's *Table of Best Grassmannian Packings* (last modified 1997), now maintained by Henry Cohn, lists values for some N in the row $m = 9$, $n = 4$. The cell $N = 10$ is blank in Sloane's table, in Sloane's coordinate files and in Cohn's table as checked on October 8, 2026. The best value implied by known packings is 2.444132, obtained by deleting 1 of the 11 four-spaces of the packing found by this search. The Rankin simplex bound for this cell is $4(9 - 4)/9 \cdot 10/9 = 2.469136$, and the value above lies below it.

3 Verification

Each subspace is given by an integer $m \times n$ basis Y (coordinates times 10^{12} , rounded). With $G = Y^\top Y$ the projection is $P = YG^{-1}Y^\top$, so for two subspaces

$$\text{tr}(P_a P_b) = \frac{\text{tr}(\text{adj}(G_a) A \text{adj}(G_b) A^\top)}{\det G_a \det G_b}, \quad A = Y_a^\top Y_b,$$

an exact rational number. No orthonormality is assumed and no floating point is used. Run

`python grassmann/verify.py grassmann/packings/m9n4N10.txt`

in <https://github.com/Ge-limin/math>.

*Found and checked by Claude, a model released by Anthropic, running on the author's laptop. The author sent the prompts.

4 How it was found

Many random starts of gradient descent on the Grassmannian against a log-sum-exp of the pairwise overlaps, sharpened over time, then a polishing stage (`grassmann/solve.py`); about 61 seconds on one laptop, with about ten such searches running at once.

A Exact value

numerator:

```
617215754695784257549478766303799037302355368953654461175073411602264226
528062742833824543782235113623836321457506739982132205977662398597936363
014277955845956288575188368509646801088282686285
```

denominator:

```
250000000000185536789985775566364390901170871180855218692945537389679459
556982445009989330419308140677446329682658929894881654228815772857025163
629310373291780363932260894067984078174763824969
```

B Coordinates

10 blocks of 9 rows with 4 integers each; divide by 10^{12} . File: `grassmann/packings/m9n4N10.txt`.

```
96357167491 256331316013 398888109158 219543954043 274979771852 -20861944677 -141594343202 -386032630481 4104387077 -159750840061 232416022252 -304296081109
-33510342768 -435713898907 168454371687 79219417373 209999985969 -225431403291 -470244181303 -130494808858 746481662 -371934188964 723942687979 135337712913
187424059453 -291573585145 -720803393963 80993802112 25326259932 -368881149690 742062489017 -265405072370 48208064 -335679509819 -109276619545 -226721660547
714564065758 -354004480180 335588438551 -292223806547 4747963651 -635703262512 -247129954250 -84934822822 582473452 29953056826 -62537085483 -393652720485
-73906487592 267580535366 169011967216 -124957490432 80611010160 -326877659576 -154118320051 580715379784 79030639 -462581829889 -523092524678 397779254894
570786916679 256973913732 -2865677721 212943904760 -169633026963 455380384926 54120497245 115061988564 1808143848 -37997325471 278870720084 -108807848436
132029942464 311849684598 -261112687013 -100035062965 287174990 255498625573 -169206646949 -459308631763 228509845 226454466124 -4716822493 505420516954
-274678099939 -88946802951 267257191364 -97304194642 70246483701 -122834759013 146788997098 -141768239313 922259221 172674620554 -38873859822 -203676353223
140438362324 539737558823 -110248670594 389665873634 11792369851 -104536239608 -116183843399 -373912303889 4755967 500550488503 224340840163 511323040992
5422671101 611521237346 269822221667 -590075156422 -184636753001 129997760249 93196398590 389022198878 172682745568 138060003411 -414718721557 185710919808
-54527745130 -258884620213 -230078981651 136248383361 542895985 450919959763 744146872040 3635851166 193222494268 -179635700306 589957988253 191775801583
9220148520 -9444032695 -264418218912 -203999836688 -15120264112 40056685957 43554526679 -589184881366 160222733946 -633007565186 269796518093 212881047234
409385004526 257862892069 -622523230171 -570628713693 106640333 -190384070216 388428546989 -206744295265 0210058928 -328475983486 108656474112 -736008104203
-294305342846 -380512344350 290485061519 -435708025760 792918337 477817179680 -1432980542 267823687500 17979609874 -548442369617 -356343005460 232259878056
111116763278 145879820159 344854614259 386794472584 86690222826 513290366018 -470220685712 121881758997 22160540622 -14231516800 -464587967879 9688523138
-214111112418 547841512311 104069534752 323249571600 4567920765 -205004692089 336686313963 88841006416 61043226711 -10730040356 464978600587 -119511014372
-143241846183 19173108598 175383264659 37977184390 53039886305 -173813480312 -147423127193 4227978417 517327072974 -296396369470 -98807551274 -106326149241
814856957822 -160840206403 414622033634 -143943190728 148865277 -8056740944 -213814179577 237968963692 05116837902 24604257799 -244975402464 -83876933008
-564647782325 -431888702983 344736562998 786133448359 20479749 -191359786603 -270882838790 -841203102706 4983032561 -429946617164 563113013126 181365937218
-393043036798 389425877801 146020670855 -362323347836 818641424 -485957389843 -368924575587 323248031743 225687643 -521552768678 -162084411611 -106027498684
259656620730 359603811999 -4658463460 -43417119077 -149063063913 109374904668 -725062714414 -93460504677 4807347117 -213703683130 126581405232 572271479276
-317113654714 -374429687008 -429155770508 -243024234942 2425927 716863534871 -67095797638 -8773876777 311591575672 453167864409 -10399598109 -267628775635
-100301778193 -32789996026 -267770109070 -410785789562 2096043 -315795717322 45534710143 -102848513807 35614991454 262863132430 -476269945275 536066100233
202915021549 -296218297154 696545265493 -39715772106 6939258797 -171291430197 278241250763 -3212713197409 5206360590 309670061543 -103582258153 138695886623
-523152894560 480056854532 87602849089 468346693004 71735566630 -35450450515 -307789251266 205448935318 225489285709 363303976001 242717312033 -65797194594
98243200035 -212542000810 -295802991633 635142757320 13309988 54020836413 14846694274 -738468339347 -443130839417 335167129690 -338222008064 203979540985
-155344109407 -166907138009 -153585422336 92296554674 1009269031 -512588448114 156914101992 92046877350 161272722071 -380581360060 -12715537105 496211185214
-339593937705 243878385933 -68393512959 111124218679 5740485672 -169913880737 -79572202322 -475460437800 8224622336 197444629045 70345371584 -282123107928
86157463377 71737995124 85058829595 501959056027 -354647753749 -646672461252 -462552907399 104372732822 12328050 66471074007 713893743750 436193772756
-681778346131 172366512800 -329389341246 -385560399973 199918552 -97396065165 -264735315149 -68638261743 71181305999 -443930819625 -272523851308 -244439166176
```